

**Bridging Ties:
A Source of Firm Heterogeneity in Competitive Capabilities**

by

Bill McEvily

Graduate School of Industrial Administration
Carnegie Mellon University
5000 Forbes Avenue
Pittsburgh, PA 15213-3890
Tel: (412) 268-4198
Fax: (412) 268-6837
email: bmcevily@andrew.cmu.edu

and

Akbar Zaheer

Carlson School of Management
University of Minnesota
835 Management & Economics Building
271 19th Avenue South
Minneapolis, MN 55455
Tel: (612) 624-5232
Fax: (612) 625-2873
email: azaheer@csom.umn.edu

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Abstract

What explains differences in firms' abilities to acquire competitive capabilities? In this paper we propose that embeddedness, in terms of firms' network of bridging ties and linkages to regional institutions, are important sources of variation in firms' acquisition of competitive capabilities. We argue that firm networks rich in bridging ties and firms' participation in regional institutions are critical vehicles for accessing new information, ideas, and opportunities leading to the acquisition of competitive capabilities in geographical clusters. Hypotheses are tested on a stratified random sample of 227 job shop manufacturers located in several regions of the US Midwest using data gathered from a mailed questionnaire. Results from structural equation modeling broadly support the embeddedness hypotheses and suggest a number of novel insights about the link between firms' networks and competitive capabilities.

Key Words: Networks, Capabilities, Geographical Clusters.

Understanding differences in competitive capabilities -- “a firm’s capacity to deploy resources...using organizational processes” (Amit and Schoemaker, 1993: 35) to achieve a competitive objective -- is a central question in the field of strategy (Nelson, 1991; Teece, Pisano, Shuen, 1992). Some of the more prominent explanations for sources of these differences include imperfections in factor markets or luck (Barney, 1986), path dependence (David, 1985), causal ambiguity and uncertain imitability (Lippman and Rumelt, 1982), and relatively immobile internal resources and capabilities (Barney, 1991). In contrast to these and other economically derived approaches to explaining firm-level differences which take an atomistic view by assuming that firms act alone, this research takes a relational perspective. More specifically, the relational perspective highlights the importance of firms’ embeddedness in networks of social, economic, and professional ties (Granovetter, 1985) to explain economic action such as alliance formation (Gulati, 1996), interfirm exchange (Uzzi, 1996b), and organizational survival (Baum and Oliver, 1992). We extend this literature by exploring how firms’ embeddedness in geographical clusters relates to their acquisition of competitive capabilities.

Recently, there has been renewed interest in the phenomenon of geographical clusters, also known as industrial districts (Piore and Sabel, 1984) or ‘hot spots’ (Pouder and St. John, 1996). Research in a number of disciplines -- including economic geography (Krugman, 1991; Scott, 1992), sociology (Lazerson, 1988), political science (Sabel, 1993), and international and strategic management (Porter, 1990) -- has documented the impressive economic growth and vitality of regions such as the ‘Third Italy’ (Pyke and Sengenberger, 1990) and Silicon Valley (Saxenian, 1994). A prominent feature of these and other similar geographical clusters is the extensive network of interfirm linkages supporting knowledge trading and collaborative innovation which occur within a context of frequent and repeated interactions.

The literature on geographical clusters argues that firms within a region tend to exhibit “...similar resources, cost structures, mental models, and competitive behavior” (Pouder and St. John, 1996: 1195). In effect, this stream of research assumes that firms in these regions are largely homogeneous and achieve similar levels of performance. However, not all firms have been equally successful at exploiting the opportunities available in geographical clusters. Whereas some firms manage to acquire and maintain the capabilities to successfully compete, others languish with obsolete skills and routines, and perform relatively poorly (Saxenian, 1994). This naturally raises the question of what explains differences in the competitive capabilities of firms within a geographical cluster.

We explain difference in firms’ competitive capabilities by viewing economic action as being embedded in firms’ network of ties, including non-market ties (Uzzi, 1996a), and the institutional context (Oliver, 1996). More specifically, the embeddedness perspective stresses that “...*networks of social relations penetrate irregularly and in differing degrees in different sectors of economic life...*” (Granovetter, 1985: 491; emphasis ours). Applied to geographical cluster, this perspective suggests that firms are each embedded in highly differentiated ways which link them to different sets of players and thereby present them with sharply distinct opportunities and constraints for acquiring competitive capabilities. Since firms vary in their ability to acquire competitive capabilities (Nelson, 1991; Teece, Pisano, Shuen, 1992), it is important to understand how a firm’s distinctive pattern of network ties relates to its acquisition of competitive capabilities. We highlight two key differentiating characteristics of firms’ networks: bridging ties and linkages to regional institutions.

Focusing on the diversity within a firm’s network, we develop the concept of *bridging ties*. By formulating a theoretical link between the firm’s internal capabilities and its external network of

bridging ties, we offer a novel approach to understanding and explaining differences in the competitive capabilities among firms in a geographical cluster. In short, we propose that networks rich in bridging ties serve as an especially powerful means for firms to gain access to new information and opportunities about enhancing their competitive capabilities.

Firms' networks in a geographical cluster are further distinguished by linkages with regional institutions supporting the geographical cluster. Regional institutions refer to local nonprofit organizations which provide collective support services to firms and include organizations such as manufacturing extension programs, technical assistance centers, university outreach programs, vocational training centers and local research institutes. Not included in this conceptualization are social and civic institutions such as the Kiwanis or the United Way.

Regional institutions facilitate the development of competitive capabilities among local firms by providing needed support services such as applied R&D, vocational training, information on new technologies and managerial practices, and market research. Beyond the tangible support services provided, regional institutions perform the specialized network role of an intermediary by organizing forums for interaction and information exchange among firms in a region. By participating in regional institutions local firms have the potential to enhance their competitive capabilities by gaining new ideas and knowledge about best practices and technologies in the region. In effect, the regional institution reduces search costs for local firms by serving as a repository of knowledge about best practices and new technologies. Although the services provided by regional institutions are available to all the firms in a geographical cluster, not all firms participate to the same degree nor do they benefit to the same extent. Therefore, a question arises as to which firms actually do utilize these services and to what extent regional institutions contribute to the development of firms' competitive capabilities.

In sum, we propose that heterogeneity in firms' networks of bridging ties and variations in their linkages to regional institutions are important sources of differences in firms' competitive capabilities in a geographical cluster. To this end, our paper addresses the following specific research questions:

- i. *How do differences in firms' network of bridging ties relate to heterogeneity in competitive capabilities among firms in a geographical cluster?*
- ii. *How are differences in firms' network of bridging ties associated with participation in regional institutions? Moreover, why do some firms use the services provided by regional institutions more extensively than others?*
- iii. *To what extent does participating in regional institutions enhance firms' competitive capabilities?*

In addressing these questions we draw on theoretical perspectives from the social network literature and highlight the network role that regional institutions play in facilitating the acquisition of competitive capabilities by small firms in geographical clusters. Based on these ideas we develop a series of hypotheses predicting which firms will participate in regional institutions and acquire competitive capabilities to varying degrees.

Following the development of hypotheses, we describe the research setting and methodology used to test our predictions. We use a stratified random sample of 227 small manufacturers located in several geographical clusters in the US Midwest to examine the validity of the hypothesized relationships. Data are collected via a mailed questionnaire sent to executives of job shop manufacturers operating in the metal-working sector. After presenting the results of our analysis, we explore the implications of the findings for theory and research. We begin with a brief discussion of knowledge sharing through interfirm networks in geographical clusters

NETWORKS AND GEOGRAPHICAL CLUSTERS

Following Porter (1990) we define a geographical cluster as a concentrated group of firms competing in the same or related industries which are linked together through vertical (buyer-supplier) or horizontal (technology, information, or other resources) relationships. Geographical clusters vary considerably in their composition of vertical and horizontal relationships. Some clusters consist of largely vertical relationships where a single dominant firm coordinates an extensive network of suppliers and subcontractors; for example Benetton in the Italian apparel industry (Heskett and Signorelli, 1982) and Toyota in the Japanese automobile industry (Asanuma, 1989). In contrast, other geographical clusters include a mixture of vertical and horizontal, competitive and cooperative, relationships. Illustrative of such clusters are the semiconductor and other electronics firms located in Silicon Valley which exchange advice and engineering solutions, information about new technologies and practices, and share knowledge through professional, social, and exchange relationships (Saxenian, 1994).

The transfer of situation-specific knowledge, and the creation and exchange of new ideas is an important attribute distinguishing geographical clusters from market coordination. Marshall recognized this point early on noting that

inventions and improvements in machinery, in processes and the general organization of the business have their merits promptly discussed; if one man starts a new idea, it is taken up by others and combined with suggestions of their own; and thus it becomes the source of new ideas (Marshall 1920: 271)

Accordingly, firms in geographical clusters enhance their competitive capabilities based, in part, on the new ideas and opportunities obtained from other economic actors in the cluster. Geographical clusters possess “the institutional capacity to continuously learn, adjust, and improve

in economic performance ... initiatives from one firm intersect with others and modify the production capabilities and opportunities for each firm” (Best, 1990: 235). Along these lines, network-based explanations emphasize that knowledge sharing among firms is enabled by high levels of trust in tight-knit industrial communities where a reputation for fair dealing and reliability is a valuable resource. In these communities, the network of relationships among firms is typically characterized as a web of dense and overlapping ties which rapidly diffuses knowledge throughout the geographical cluster.

However, taken to the logical extreme, overlapping networks may actually constrain, rather than enable, access to knowledge about innovative practices and techniques. For instance, it is unclear how new ideas and information would enter an entirely closed social system of strong ties. Rather, it seems unlikely that all firms in a geographical cluster maintain an identical pattern of linkages. Since interfirm relationships are frequently formed as a result of commonalities – such as family ties, membership in civic groups and social clubs, and new business spin-offs from parent corporations or universities -- firm’s networks are more dense and overlapping with those of other firms with whom they have a shared heritage. At the same time, firms’ networks are likely to be quite distinct from those of other firms with whom they do not share a common history. Viewed this way, geographical clusters contain a host of “clumps” or sub-clusters of thick and multiplex interfirm networks in certain areas, while the areas in-between sub-clusters are sparser with fewer linkages across clumps. This alternative perspective of geographical clusters suggests that each firm maintains a unique and idiosyncratic pattern of network linkages and consequently are differentially exposed to new knowledge and ideas leading to the acquisition of competitive capabilities. In the next section, we develop a conceptual framework for explaining the theoretical relationship between a firm’s network structure and acquisition of competitive capabilities.

THEORY AND HYPOTHESES

In this section, we employ the embeddedness perspective to develop the idea of bridging ties and to relate this concept to firms' competitive capabilities and participation in regional institutions. We argue that a firm's network strategy is manifested in its choice of contacts, and more importantly its overall pattern of bridging ties with other economic actors.

Bridging Ties

We define bridging ties as those that link a focal firm to contacts which primarily operate in diverse economic, professional, and social circles from those of the firm. We theorize that bridging ties connect a focal firm to sources of information and opportunities which are not readily available from its other network contacts. In geographical clusters, firms which act as connectors between sub-clusters represent bridging ties. The concept of bridging ties has its basis in theories of social networks which have argued that new information is predominantly obtained through 'weak ties' (Granovetter, 1973) and 'structural holes' (Burt, 1992). Granovetter posited that it is through casual acquaintances, or weak ties, that new information is obtained, rather than through strong ties, or close personal friends. The weak tie thesis is based on the reasoning that strong ties tend to result in a dense cluster of actors who are all mutually connected to each other. In other words, the individuals that an actor has strong ties with also tend to have strong ties with each other. Since this sub-cluster of strongly connected actors is likely to interact with each other frequently, much of the information circulating in this social system is *redundant*.

Conversely, weak ties are often links with actors who move in different social circles than the focal actor and are therefore exposed to a different set of information and opportunities. The

‘strength of weak ties’ argument rests on the assertion that opportunities are discovered through weak ties because these serve as bridges to new and different information. For example, in a study of re-employment, Granovetter (1973) found that an individual’s acquaintances, people with whom only sporadic contact had been maintained, were critical to discovering new employment opportunities. In his study of re-employment, weak ties were conceptualized as the relative infrequency of interaction between the focal actor and contacts.

However, infrequency of interaction alone may not be a sufficient condition for discovering opportunities if an actor’s weak tie contacts are either connected to each other or connected to the actor’s strong tie contacts. In this vein, Burt (1992) extends the strength of weak ties argument by asserting that it is not so much the strength or weakness of a tie that determines its information potential, but rather whether a *structural hole* exists between a focal actor’s contacts (i.e. the contact is disconnected from other contacts in the focal firm’s network). Put differently, the key underlying causal mechanism that determines whether a tie will provide access to new information and opportunities is the extent to which it is non-redundant or, in Burt’s terminology, whether a tie spans a structural hole.

As Figure 1 illustrates, the weak tie between Actor 1 and 2 serves as a bridge by connecting the focal actor (Actor 1) to a contact (Actor 2) disconnected from the focal actor’s other direct contacts. However, it can also be seen from the linkage between Actor 1 and Actor 3 that strong ties may equally serve as a bridge since the connection is non-redundant. The strength of weak ties and structural holes arguments offer compelling theoretical explanations for the discovery of new information and opportunities. However, a question remains as to whether non-redundancy or infrequency of interaction offers a more empirically accurate explanation.

We conceptualize the notion of bridging ties as consisting of three elements, of which two - non-redundancy and infrequency of interaction -- are drawn from the literature reviewed above. The firm's advice network represents an appropriate context for defining and operationalizing the bridging ties constructs since it serves as a key source of new information, ideas and opportunities. More specifically, we define *non-redundancy* as the extent to which the contacts in a focal actor's advice network are not linked to one another. *Infrequency of interaction* refers to the rarity with which a focal actor converses with the contacts in its advice network. To these two we add a third, geographic dispersion, which is particularly relevant to geographical clusters.

Geographic dispersion in the advice network is a proxy for the lack of opportunity for face-to-face interaction among advisors. In our view, a dispersed advice network comprises contacts located spatially far enough apart to impede routine face-to-face interaction. With a geographically scattered advice network, contacts interact less readily with the focal firm, as well as with each other, and therefore provide non-redundant information.

We view non-redundancy, infrequency of interaction, and geographic dispersion in the advice network as related, but distinct concepts. Each in a different way captures the potential, or the lack thereof, for bridging ties. In particular, we argue that bridging ties exist where the advice network consists of high non-redundancy, infrequency of interaction, and geographic dispersion.

Bridging Ties and Competitive Capabilities. We propose that firms with an advice network rich in bridging ties will acquire competitive capabilities to a greater extent than firms with an advice network lacking in bridging ties. More specifically, advisors will have access to similar information about competitive capabilities when they are all tied to each other. In other words, little fresh news circulates in a highly redundant social system. In contrast, when a firm's advisors are complete strangers to one another, they operate in different social circles and know of

different opportunities to innovate and develop competitive capabilities. Firms that maintain linkages to diverse information sources gain access to novel information and learn about competitive capabilities to a greater degree than firms without such ties. By maintaining such a network of diverse and non-overlapping ties, a firm increases its chances of gaining access to new ideas and information about competitive capabilities. This leads to the first hypothesis,

H1: Non-redundancy in the firm's advice network will be positively related to its acquisition of competitive capabilities.

Similarly, an advice network composed of contacts that the focal firm interacts with frequently is less likely to be of the bridging type. Based on Granovetter's (1973) assertion that the contacts that an actor interacts with frequently (i.e. strong ties) tend to interact with each other frequently as well, we should observe a dense cluster of mutually connected advisors when a firm interacts with its advisors frequently. This suggests that the less a firm interacts with the contacts in its advice network, the less a firm's advisors interact with each other. Further, where a firm's advisors interact with each other infrequently the firm should gain access to new information about competitive capabilities. These ideas are summarized in the following hypothesis,

H2: Infrequency of interaction with the firm's advice network will be positively related to its acquisition of competitive capabilities.

An advice network rich in bridging ties also exists when advisors are geographically dispersed. More specifically, with a geographically scattered advice network face-to-face interaction among contacts diminishes. In contrast, a densely clustered advice network around the firm provides greater opportunity for a firm's contacts to interact and contains more overlapping

ties and redundant information. In other words, as the opportunity for face-to-face interaction declines a firm's contacts access less similar information. Hence, the diversity of information obtained by a firm increases with the geographic dispersion of the firm's advice network which enhances the acquisition of competitive capabilities.

H3: Geographic dispersion in the firm's advice network will be positively related to its acquisition of competitive capabilities.

Participation in Regional Institutions and Capabilities. The connection between firms' participation in regional institutions and their acquisition of competitive capabilities derives from the specialized network role of an intermediary that regional institutions perform. As intermediaries, regional institutions facilitate the acquisition of competitive capabilities by compiling and disseminating knowledge, convening forums for interaction, and reducing search costs.

Beyond providing specific support services -- such as vocational training, market research, and applied R&D -- to local firms, regional institutions also act as repositories for knowledge and opportunities about competitive capabilities. Due to the fact that regional institutions interact with a relatively large number of firms in the geographical cluster, they are exposed to a wide variety of solutions to organizational challenges typically faced by the firms in a region. Based on their broad experience of observing others who have dealt with similar problems, regional institutions compile and disseminate ready-to-use summaries about competitive capabilities, competencies, and routines (Suchman, 1994). In effect, regional institutions facilitate managerial innovation by providing access to information and resources for acquiring new, and extending existing, capabilities.

Another aspect of regional institutions performing an intermediary role derives from their capacity to convene forums which provide a means of “...linking speakers and audiences wherein meaning is created and communicated through discussion, debate, or deliberation” (Bryson and Crosby, 1992: 92). In effect, forums provide an arena within which knowledge sharing among firms in a network and the identification of sources of capabilities (e.g. potential employees, alliance partners, and consultants) can more readily take place. By condensing networks in time and space, the forums organized by regional institutions create and sustain regular patterns of interaction among firms in a geographical cluster (Nohria, 1992; Saxenian, 1994; self cite).

Regional institutions also mitigate search costs associated with locating external sources of knowledge and specialized expertise critical to the acquisition of competitive capabilities. The search and transactional economies generated by intermediaries stem from their maintaining an extensive network of ties to different parts of a social system which allows individual members of a network to “...relieve themselves of a heavy burden and free up their own time and energy to engage in other activities” (Galaskiewicz, 1985: 113). Therefore, an actor can maintain a single connection with the intermediary who specializes in providing access to and information about competitive capabilities, rather than maintaining numerous ties throughout the network.

A key insight developed in the social networks literature is the idea that intermediaries reduce search and other transaction costs associated with exchange (Galaskiewicz, 1985). Intermediaries serve as a go-between for potential exchange partners who are otherwise disconnected (Aldrich and Whetten, 1981). In effect, intermediaries bridge the social gaps in a network by “linking persons having complementary interests, transferring information, and otherwise facilitating the interests of persons not directly connected to one another” (Aldrich and Zimmer, 1986: 16). Within the context of geographical clusters this suggest that rather than all

firms being tied to one another, each can maintain a single connection with the regional institution which specializes in providing access to and information about potential exchange partners.

In summary, for participating firms, regional institutions fulfill an intermediary network role, by serving as a repository of knowledge, organizing forums, and reducing search costs.

Based on these ideas, we hypothesize

H4: Participation in regional institutions is positively related to the firm's acquisition of competitive capabilities.

Bridging Ties and Participation in Regional Institutions. In spite of the seemingly obvious advantages of participating in regional institutions, firms are not always inclined to utilize their services. In part this reluctance is due to the fact that participation requires firms to simultaneously balance the opposing tensions of cooperating and competing with other participants. While regional institutions do provide firms with opportunities to learn about new capabilities, participants are also exposed to the risk that proprietary information about their own competitive advantage will be revealed to rival firms. Firms may hesitate to participate in regional institutions because they are concerned that they may lose more than they gain.

Ambivalence about participating in a regional institution stems from the fact that firms often need to share a certain amount of information about their own operations with the regional institution. In order to receive assistance with an organizational problem, a firm must provide some information about its current operations so that the issue can be analyzed and diagnosed. In many cases this involves a visit to and inspection of the firm's operations. This presents the potential for exposing proprietary and sensitive information about the firm's source of competitive advantage, some of which may be unrelated to the specific problem at hand.

Thus, whatever knowledge, expertise, or skills that are developed in the process of resolving a specific organizational challenge confronted by a firm become part of the regional institution's 'stock of knowledge'. Since this knowledge can then be shared with the firm's direct competitors, there is limited potential for such learning to be the basis of a sustained competitive advantage. Viewed this way, using the specialized services available from a regional institution is a way of indirectly helping a firm's competitors resolve similar problems.

Consequently, participating in a regional institution may be viewed as posing a threat to a firm's strategic advantage since whatever knowledge, skills, or expertise that are acquired through a regional institution become part of the stock of public knowledge. As a quasi-public entity, the regional institution seeks to spread such knowledge as widely as possible. Thus, firms which have acquired capabilities through their bridging ties would be less inclined to use the specialized services available from a regional institution. Rather these firms would want to keep proprietary whatever knowledge, capabilities, and skills they have acquired. As before, bridging ties are conceptualized as non-redundancy, interpersonal closeness, and geographic proximity. Based on these arguments we predict that,

H5: Non-redundancy in the firm's advice network will be negatively related to its participation in regional institutions.

H6: Infrequency of interaction in the firm's advice network will be negatively related to its participation in regional institutions.

H7: Geographic dispersion in the firm's advice network will be negatively related to its participation in regional institutions.

RESEARCH METHODS

Research Setting

The context for this research project are the firm clusters located in two states of the US Midwest. These clusters are served by industrial extension centers affiliated with a national system of such centers which, for the purposes of this study, represent regional institutions. The recent emergence and growth of a nationwide infrastructure of regionally-based industrial extension centers provides an ideal opportunity for exploring research questions relating to the acquisition of competitive capabilities and participation in regional institutions.

Beginning in the late 1980s, a new set of institutional arrangements began to appear in the U.S. with a specific focus on promoting the deployment of new manufacturing technology, production techniques, and business practices among small manufacturers. This new public infrastructure was embodied in a system of regional industrial extension centers, managed by National Institute of Standards and Technology (NIST) which assists small manufacturers with productivity enhancement through upgrading their manufacturing technology and improving their organizational practices.

The cornerstone of NIST's efforts to improve the competitiveness of small manufacturers is its Manufacturing Extension Partnership (MEP) initiative. The regionally-based MEP centers provide a common set of in-house services such as information 'hot lines', training courses, workshops and conferences, equipment demonstrations, supplier certification, and manufacturing assessments. Each center also coordinates a staff of field agents performing a range of outreach activities such as reconfiguration of existing manufacturing operations, selection and installation of new equipment, and identification and implementation of new management practices.

Research Design and Data Collection

This study uses a stratified random sample of firms, based on participation in a regional industrial extension center, to test the hypotheses. The sampling frame consists of all job shop manufacturers located in several regions of two Midwestern states and operating in the Metal Working sector (i.e. Electroplating, Coating and Painting, Printed Circuit Board Manufacturing, Screw Machining, Stamping, Sheet Metal Fabrication, and Machining). These regions were selected because each operates an office of the state MEP. The Metal Working industries were chosen to obtain roughly equivalent proportions of firms participating in each region's industrial extension center. In addition, these industries are all predominantly populated by small and mid-sized firms and tend to rely on similar production technologies.

A primary dataset was constructed using a mailed questionnaire. Officials from the MEPs in both states compiled and made available a list of participating and non-participating firms in each of the industries being studied. From this list, all participating firms in each industry and an equal number of randomly selected non-participating firms in the same industry and state were extracted. A preliminary version of the survey instrument was pretested amongst a group of local executives of job shop manufacturing companies from an industry not among those identified above. The questionnaire was administered to 22 job shop manufacturers, including a roughly equal number of participants and non-participants. Feedback from these executives was incorporated into a revised version of the survey instrument along with comments and suggestions from industry experts, officials from the MEPs, and several colleagues knowledgeable in survey design.

A copy of the final questionnaire was mailed to a total of 1,000 Chief Executive Officers or Presidents who are best able to respond to questions about organizational and strategic issues relating to their firm. This is consistent with the approach of selecting key informants

knowledgeable about organizational matters by virtue of their position (John and Weitz, 1988). The sample included approximately 500 participants and 500 non-participants across both states and all Metal Working industries. We also implemented Dilman's (1978) techniques for maximizing response rate. Most importantly, in addition to the initial mailing of the survey, extensive follow-up communications was carried out including the following sequential steps: (1) sending a reminder/thank you postcard two weeks after the first round of mailing, (2) placing a telephone call to non-respondents, and (3) sending a second round of mailings to any remaining non-respondents.

A total of 309 executives responded to the request for information about their company and returned a questionnaire. This equates to approximately 31% of the original 1,000 firms surveyed. However, through the follow-up telephone calls it was learned that 178 firms were not eligible to participate in the study¹ and we eliminated these firms from our initial sampling frame. Therefore, the actual response rate equals 38% (i.e. 309/822) of the firms that were eligible to respond. Complete responses are available from 227 of these firms.

Testing for Non-Response Bias. Although the overall response rate of 38% is quite reasonable, particularly given the reluctance of small manufacturers to respond to mailed surveys, the possibility remains that the sample of firms responding to the questionnaire systematically differs from the remainder of the population. We addressed the potential for non-response bias by comparing certain key attributes of respondents to those of a group of 50 randomly selected non-respondents with firm size (i.e. number of employees) and annual sales data from one of the MEPS. T-tests revealed no significant differences between the mean size ($t = -1.29$) and the mean sales ($t = -1.83$) of respondents and non-respondents, although the near significance of the difference in the

mean sales suggests the possibility of some bias in our sample toward firms with higher sales. Nevertheless, these tests provide some assurance that the sample of firms responding to the questionnaire is representative of the broader population surveyed.

Operational Measures

This study uses a multi-measure approach to operationalize the theoretical constructs. Whenever possible, validated measurement instruments available from extant research are used to operationalize the theoretical constructs. Several instruments were modified to make them more suitable for the present research setting. Table 1 presents the details of the measurement instruments and scales used to operationalize our theoretical constructs. The Cronbach α reliabilities for each construct are also reported in Table 1, which are all at or above the recommended value of .70 (Nunnally, 1978) with the exception of the quality management construct which is marginal at .61. Below, the specific details of the development and operationalization of the theoretical constructs are discussed.

Acquisition of Competitive capabilities. This construct is based on the ‘assimilation of innovation’ scale developed by Meyer and Goes (1988). These authors argue that innovation adoption in the organizational context is not simply a discrete event that is carried out instantaneously by individual decision makers. Rather, an innovation presents organizations with an adoption opportunity and triggers a set of formal and informal decision processes that may end in awareness, evaluation, adoption, utilization, and institutionalization.

In order to develop a variable for statistical analysis, we created a 7-point scale which captures three primary decision-making stages associated with the adoption of technology:

¹ Firms deemed ineligible to be include in our study include those that were not job shop manufacturers, had gone out

knowledge-awareness, evaluation-choice, adoption-implementation. Since we view the acquisition of new competitive capabilities as analogous to the adoption of technological innovation the wording of the Meyer and Goes (1988) measurement scale was adapted. Specifically, the modified scale refers to competitive capabilities, rather than technological innovations, and focuses on small, rather than large, organizations.

Three competitive capabilities are evaluated using the acquisition scale: pollution prevention, competitive scanning, and quality management. Following Amit and Schoemaker (1993) we focus on industry-specific competitive capabilities that have a significant bearing on firms' capacity to undertake a particular activity that is intended to lead to the attainment of a strategic objective. These three competitive capabilities were selected because industry experts identified them as among the most critical activities affecting firm performance in the selected industries.

Pollution Prevention. Pollution prevention capabilities have become salient with recent amendments to environmental regulations, such as the Clean Water Act, governing the use and emission of hazardous materials which are integral to the production process of industries in the Metal Working sector. Pollution prevention emphasizes the judicious use of resources through source reduction, energy efficiency, reuse of scrap materials during production, and reduced releases of hazardous or toxic materials (U.S. Environmental Protection Agency, 1990). Product and process changes are two common methods used to reduce or eliminate the creation of pollutants or wastes at the source. Product changes include altering the composition or use of intermediate or end products. Process changes are manufacturing modifications that affect the amount of waste generated as a result of the way a product is produced. Pollution prevention is

operationalized with three items measuring various product and process changes (see Pollution Prevention instrument in Table 1). These items were adapted from a national survey of pollution prevention conducted in one of the job shop industries selected for this study (Cushnie, 1994).

Competitive Scanning. Given the need to monitor competitors' actions closely in an environment of low switching costs and lack of customer loyalty, competitive scanning is critical to firm competitiveness in these Metal Working industries. Central to the notion of sustainable competitive advantage is the idea that firms strive to insulate themselves from the uncertainties of competitive interaction by establishing a unique position relative to rivals (Grant, 1995). Consequently, strategy formulation implicitly entails scanning the external environment for information about competitors in order to identify threats and opportunities (Andrews, 1980; Aguilar, 1967). The concept competitive scanning is operationalized in this study using three items that tap into the practices a firm uses to gather information about rivals (see Competitive Scanning instrument in Table 1). These items are based on the measurement instrument created and validated by Miller (1987).

Quality Management. Quality management has increasingly become a prerequisite for doing business in Metal Working industries which are part of supply chains supporting the automobile and other global industries. Although total quality management (TQM) has come to mean different things to different people, the elimination of production defects through continuous improvement of all processes is a common feature of most definitions (Hackman and Wageman, 1995). A core principle guiding the management of quality improvement is the analysis of variability in processes and outcomes through systematic data collection and the use of statistics (Deming, 1986; Ishikawa, 1985; Juran, 1974). In this study, TQM is measured with three items capturing the use of statistical process control charts to provide operators with feedback which

allows them to base their actions on the variability of the manufacturing process (see Quality Management instrument in Table 1). These items are based on the measurement instrument developed by Flynn, Sakakibara, and Schroeder (1995: 1327).

Participation in Regional Institution. This construct indicates the extent to which a firm has utilized the services available from a regional industrial extension center. Analogous to the acquisition of competitive capabilities, participation in a regional institution is not a discrete event, but rather represents an opportunity for a small manufacturer to access information and ideas.

Similar to the acquisition scale, a 7-point scale was developed to measure participation and captures three primary decision-making stages: knowledge-awareness, evaluation-choice, and utilization. Selecting and implementing new management practices or new technologies are characteristic of participation and is measured with a three item scale capturing the use of these service categories (see Participation instrument in Table 1).

Non-redundancy. To operationalize non-redundancy, this study uses an ego-centered network measure based on an instrument developed by Aldrich, Rosen, and Woodward (1986) specifically designed for use in the small firm context. This instrument asks respondents (ego) to identify the five most important external sources of advice (alters) relied upon and to report the extent to which these five sources know each other (see Non-redundancy instrument in Table 1).

Using this matrix a non-redundancy score is computed as,

$$\text{Non-redundancy} = (\text{Potential Ties} - \text{Actual Ties}) / \text{Number of Advisors}$$

where,

Potential Ties = the maximum number ties that could exist among advisors, or $n(n-1)/2$; where, n is the total number of advisors listed (0 to 5)

Actual Ties = the number of advisors who know each other

Number of Advisors = the total number of advisors listed (0 to 5).

This equation defines non-redundancy as a ratio of non-redundant ties per advisor. The values for this variable range from zero to two with low numbers indicating low non-redundancy and high numbers reflecting a high degree of non-redundancy. In the case where all advisors know each other, non-redundancy equals zero (i.e. there is no non-redundancy, but complete redundancy). In contrast, when no advisors know each other non-redundancy reaches its maximum of two reflecting a network of ties which is entirely non-overlapping. In other words, a low percentage of advisors that know each other is an indication that a firm's network is rich in bridging ties.

Infrequency of Interaction. This variable measures the rarity of interaction with advisors in terms of the number of conversations per month (see Infrequency of Interaction instrument in Table 1). Respondents were asked to report the average number of conversation held per month with each of the advisors listed. Using this information, an overall average infrequency of interaction score for each respondent is computed as,

$$\text{Infrequency of Interaction}^2 = \frac{1}{\sqrt{\text{mean (conversations per month)}}}$$

Geographic Dispersion. In order to operationalize this variable respondents were asked to indicate how far away each advisor is located in terms of travel time by car (see Geographic Dispersion instrument in Table 1). This data was used to compute an aggregate geographic dispersion score for each respondent based on the average distance to an advisor as follows,

$$\text{Geographic Dispersion}^3 = \sqrt{\text{mean (distance)}}$$

² A square root transformation is used to normalize these data.

³ A square root transformation is used to normalize these data.

Control Variable. We include firm size in terms of the total number of employees (as standardized z-scores) as a control since larger firms are more likely to acquire competitive capabilities at a greater level and participate in regional institutions to a greater degree.

Construct Validity

A measure of a construct is valid to the extent that it actually measures what it purports to measure (Carmine and Zeller, 1979). The logic of construct validity suggests that multiple indicators of the same theoretical construct should be positively and strongly related. In particular, convergent validity refers to “...the degree to which multiple attempts to measure the same concept by different methods are in agreement” (Philips, 1981: 399; Campbell and Fiske, 1959). The degree to which two theoretical constructs differ from each other indicates discriminant validity.

In general, we use previously validated instruments, adapted to our research context, to measure the constructs in this study. However, despite the importance of accurate assessment of network ties among alters by ego for the validity of ego-centered network constructs, these measures have received little attention (Wasserman and Faust, 1994). For this reason, we evaluated the validity of the bridging ties constructs using data gathered from a second knowledgeable respondent who could provide an accurate report on the network data.

Validity of Network Constructs. The ego-centered network data described earlier relies on a surveying strategy which asks a focal actor (ego) to identify a set of alters who are connected to ego, and then report on the network of ties among these alters. The validity of the network constructs based on this ego-centered network data obviously hinges on ego’s ability to accurately assess the ties between pairs of alters.

In order to validate the constructs based on the ego-centered network data gathered, this study surveyed the alters of 40 randomly selected firms of which 21 agreed to provide access to their contacts. All of the 85 contacts identified by the 21 respondents were called and agreed to confirm their ties with the other alters listed by ego (i.e. the focal firm). We find that 72% of the time ego's assessment of the tie between a pair of alters corresponds with *both* alters' report of the tie between themselves. Further, 86% of the time, ego's report of the tie between a pair of alters is in agreement with at least one of the alters. In other words, we can be fairly sure that, on average, these measures are no less than 72% accurate, which is well within the range of generally acceptable measurement accuracy. Based on this analyses, it seems reasonable to conclude that the validity of the ego-centered network measures are acceptable.

Discriminant Validity. The discriminant validity of two constructs can be assessed by demonstrating that the correlation between a pair of constructs is significantly different than unity. We test discriminant validity by comparing a structural equation model with the correlation between two constructs constrained to equal one with an unconstrained model. A significantly lower χ^2 value for the unconstrained model supports the discriminant validity criterion. As Table 3 indicates, the three bridging ties constructs as well as the three competitive capabilities constructs all exhibit satisfactory discriminant validity.

ANALYSIS AND RESULTS

In order to test the hypotheses developed earlier, we use the maximum likelihood estimation procedure in LISREL 7 (Jöreskog and Sörbom, 1993) to specify a structural equation model. Rather than assuming that constructs are measured without error, LISREL explicitly models the measurement error in the indicators. This feature is highly relevant to this study since it

relies heavily on psychometric measurement instruments. LISREL also provides the capability of simultaneously estimating a system of structural equations. This is particularly important for the present study since we test a partial-mediation model (Venkatraman, 1989) which predicts both direct and indirect effects, through participation in regional institutions, of the independent latent variables on the acquisition of competitive capabilities.

The structural equation model estimates a series of path coefficients reflecting the relationships specified among the latent variables. The Beta coefficient (β) indicates the structural path among dependent (endogenous) latent variables, which in this case is the link between participation in regional institutions and acquisition of competitive capabilities. The Gamma coefficient (γ) represents the structural paths between the independent (exogenous) latent variables and the dependent (endogenous) latent variables. The sign and statistical significance of the Beta and Gamma path coefficients serve as a test of the hypothesized relationships. The results reported for the structural equation model presented below include the path coefficients and t-values (shown in parentheses) corresponding to each hypothesized relationship, and a series of standard fit indices⁴ reflecting the degree of overall fit between the actual and predicted covariances among variables of a model. The χ^2 statistic tests the correspondence between the model and the underlying data. A non-significant χ^2 value indicates that the model is not significantly different from the underlying data. Descriptive statistics and zero-order correlations among the constructs are reported in Table 2.

⁴ Four fit indices are reported for each of the models estimated. The goodness of fit index (GFI) indicates the relative amount of variance and covariance jointly explained by the model. The adjusted goodness of fit index (AGFI) is similar to GFI, but adjusts for the number of degrees of freedom in the model. The normed fit index (NFI) (Bentler and Bonett, 1980) represents the point at which the model being evaluated falls on a scale from a null model (specifying mutual independence among indicators) to a perfect fit. The comparative fit index (CFI) (Bentler, 1990) is the same as the NFI, but corrects for small sample size by subtracting the degrees of freedom from their corresponding chi-square

The structural equation model specifies the relationship between the three bridging ties constructs and the three acquisition of competitive capabilities constructs, mediated by participation in regional industrial extension centers and controlling for firm size (see Figure 2). The structural equation model estimated demonstrated a χ^2 (83 df) value of 93.68 ($p = .195$). All fit indices are above 0.9 indicating a satisfactory fit of the model with the data. Below the results for each hypothesis are presented.

H1. The first hypothesis predicting a positive relationship between non-redundancy and competitive capabilities is *partially supported*. Two of the three competitive capabilities are significantly associated with non-redundancy in the advice network. More specifically, while the link between non-redundancy and pollution prevention capabilities is positive and statistically significant ($\gamma_{21} = 0.215$, $t = 2.433$, $p < .05$), and the relationship between non-redundancy and competitive scanning capabilities is positive and marginally significant ($\gamma_{31} = 0.157$, $t = 1.925$, $p < .10$), the link to quality management capabilities is positive but not significant ($\gamma_{41} = 0.066$, $t = 0.785$, n.s.).

H2. The hypothesized positive relationship between infrequency of interaction and competitive capabilities is *not supported*. Although infrequency of interaction is significantly associated with one of the three competitive capabilities studied, competitive scanning, the sign is negative rather than positive as predicted ($\gamma_{32} = -0.146$, $t = -1.836$, $p < .10$). The relationship between infrequency of interaction and pollution prevention capabilities is also negative, although not statistically significant ($\gamma_{22} = -0.009$, $t = -0.106$, n.s.), while the link with quality management capabilities is positive, but not statistically significant ($\gamma_{42} = 0.019$, $t = 0.228$, n.s.).

values. Each index ranges from zero to 1.00, with values closer to 1.00 indicating a good fit. A commonly accepted rule of thumb is that a fit index should be greater than .90.

H3. The predicted positive relationship between geographic dispersion and competitive capabilities is *weakly supported*. There is a positive and marginally significant relationship between geographic dispersion and competitive scanning ($\gamma_{33} = 0.138$, $t = 1.742$, $p < .10$), however the data reveal non-significant statistical relationships for both the geographic dispersion to pollution prevention link ($\gamma_{23} = -0.061$, $t = -0.714$, n.s.) and the geographic dispersion to quality management relationship ($\gamma_{43} = 0.060$, $t = 0.733$, n.s.).

H4. The fourth hypothesis positively relating participation in regional institutions to the assimilation of competitive capabilities is *partially supported* by the data. Indeed, we observe positive and statistically significant relationships for the link between participation and pollution prevention capabilities ($\beta_{21} = 0.190$, $t = 2.135$, $p < .05$) and for the link between participation to quality management capabilities ($\beta_{41} = 0.271$, $t = 3.116$, $p < .01$). However, the relationship between participation and competitive scanning capabilities is positive, though not statistically significant ($\beta_{31} = 0.126$, $t = 1.546$, n.s.).

H5. The model provided *strong support* for the fifth hypothesis which predicted a negative relationship between non-redundancy and participation in regional institutions. As predicted, the non-redundancy to participation link is both negative and statistically significant ($\gamma_{11} = -0.175$, $t = -2.087$, $p < .05$).

H6. The prediction that infrequency of interaction and participation in regional institutions would be negatively related is *not supported*. While negative, the relationship is and non-significant ($\gamma_{12} = -0.083$, $t = -1.000$, n.s.).

H7. The hypothesized relationship predicting a negative relationship between geographic dispersion and participation is also *not supported*. Rather the link between these two variables is positive, though not statistically significant ($\gamma_{13} = 0.031$, $t = 0.376$, n.s.).

Before discussing our findings it is important to mention some limitations. First, due to the cross-sectional nature of our research design we are cautious in inferring the direction of causality among the key constructs. Although we have presented the findings in a manner which implies a certain causal chain, it is possible that the direction of causality may be reversed in some cases. We believe it is more likely, however, that the constructs are related to each other in a reciprocal fashion. For instance, firms lacking bridging ties may participate in regional institutions which in turn serve as a source of bridging ties for the firm, thereby causing the firm to subsequently by-pass the regional institution. Despite this limitation, we are reasonably confident of our interpretations which are based on strong theoretical rationale and due to the triangulation from the extensive field interviews we have conducted. A further challenge we confronted was operationalizing the bridging tie concept. Because we limited ourselves to five advisors and the links among them, we may not have captured the full effects of non-redundancy in the extended network. We justify this approach by pointing out that our research strategy emphasized a multi-dimensional measurement of bridging ties which was theoretically derived. Further, network operationalization was intentionally developed with the aim of creating a parsimonious bridging tie measure.

DISCUSSION

In contrast to firm-centric approaches to explaining firm heterogeneity, this research adopts an embeddedness perspective to uncover network-based differences in competitive capabilities across firms. Our central premise is that a firm's network of interfirm ties and institutional linkages

expose it to new ideas, information, and opportunities about competitive capabilities. In effect, the distinctiveness in firms' networks, and the information access they provide, have important implications for firms' capacity to compete.

The results demonstrate that the structure of firms' networks of bridging ties influences both the acquisition of competitive capabilities and participation in regional institutions. In geographical clusters, keeping on top of developments is better achieved by maintaining a network of advisors who are non-overlapping and disconnected. This implies that the firm's configuration of linkages with other actors in the cluster are an important vehicle through which its skills, competencies, and routines are continually upgraded, refreshed, and renewed. The link between networks and capabilities contributes a valuable insight to the literature on competitive capabilities which has yet to formulate a theory for the source of competitive capabilities. In particular, while the strategic management literature has explained performance differences in terms of resources and capabilities, there is little by way of theory explaining how firms identify, obtain, and develop competitive capabilities. Some previous work in this area has suggested that capabilities derive from external linkages such as alliances (Hamel, 1993) and networks of information (Zaheer and Zaheer, 1997) or learning (Powell, Koput, and Smith-Doerr, 1996). We extend this stream of research by elaborating how firms' pattern of bridging ties and the nature of their linkages with regional institutions both create competitive capabilities.

We do not mean to imply that interfirm networks entirely explain differences in firms' competitive capabilities, which are undoubtedly also the outcome of other factors including immobile resources, causal ambiguity, luck, and path dependence. However, we do maintain that firms' embeddedness in networks of social, economic, and professional ties is an important, but somewhat neglected perspective in strategy research. The embeddedness perspective suggests that

the acquisition of competitive capabilities does not take place in a vacuum. Rather, competitive capabilities are acquired through a relational process of learning which entails identifying, evaluating, and internalizing new ideas, practices, technologies, and processes from other actors.

However, this line of reasoning obviously raises the question of why other actors would be willing to share information about their competitive capabilities with rival firms. Consistent with the notion of embeddedness, we suggest that bridging ties are not necessarily only connections between a focal firm and its competitors. Rather, bridging ties also commonly exist with symbiotic partners – such as customers, suppliers, professional service providers (e.g. bankers, accountants, and consultants), investors, and firms in related industries – which all maintain a considerable interest in the competitive success of the focal firm. Moreover, such bridging ties with a firm's main advisors are most likely to exist with trusted actors which provide reliable and valuable information. The implication for network theory is that rather than focusing on the strength or weakness of a tie, new ideas, information and opportunities can be sourced through contacts which are simultaneously non-redundant and trusted. For strategy research, because high trust relations penetrate irregularly and in differing degrees, bridging ties offer an important explanation for the differences among firms' competitive capabilities.

Our research contributes to the network literature through our innovative conceptualization of the bridging ties constructs in terms of non-redundancy, infrequency of interaction, and geographic dispersion. By relating each of the bridging ties constructs to competitive capabilities and to participation we test the explanatory power of the underlying theories (Granovetter, 1973; Burt, 1992). The results indicate that non-redundancy in the firm's advice network explains the acquisition of capabilities and participation in regional institutions, tentatively supporting the structural holes thesis, while infrequency of interaction and geographic dispersion show virtually no

effects. The implication of this pattern of findings is that the diversity of information sources is best reflected in the lack of overlap among contacts, rather than the intensity of interaction with advisors.

However, before concluding that the weak ties proposition is not supported we encourage future research to test other facets of the theory. For instance, a study mapping the entire network of advisors would allow for distinctions between weak ties with recluses versus weak ties with strategically-situated actors to be examined.

This research further demonstrates that participation in regional institutions mediates the link between bridging ties and the acquisition of competitive capabilities. On the one hand, the findings indicate that firms with an advice network rich in bridging ties tend to acquire and develop competitive capabilities independent of the regional institution. Conversely, those firms with an advice network limited in bridging ties participate to a greater extent in regional institutions. We interpret this finding as suggesting that a participating firm's link with a regional institution may act as a kind of 'surrogate-tie' by serving as a functional substitute for its own cluster of bridging ties. In effect, a regional institution acts to condense networks in time and space, thereby allowing a firm's link to it -- the surrogate-tie -- to serve as a conduit for a wide range of information and opportunities.

The fact that the regional institutions in this study appear to be effective at promoting the development of competitive capabilities among firms begs the question of why all firms do not participate and take advantage of the service provided. We suggest that the reason some firms do not participate is because they are exposed to information and opportunities through their own non-redundant contacts. Indeed, the challenge of balancing the requirements of simultaneously cooperating while competing with other firms in the geographical cluster was noted by one executive we spoke with who commented:

I think I find [the regional institution] more useful to our association as a group. I'm not saying there'd never be anything we would work with them on, but the problem in working with them is whatever we do, and do develop, is now public knowledge...anything they came in here and did would be common knowledge and wouldn't be proprietary to [our company].

What this implies is that firms that already have well-developed capabilities may be unwilling to participate in the regional institution for fear of eroding their competitive advantage. At the same time, firms that are far behind in terms of the sophistication of their capabilities may completely lack the absorptive capacity (Cohen and Levinthal, 1990) that would enable them to graft capabilities into their organizations from the regional institution. This reasoning suggests an inverted-U shaped relationship between participation in regional institutions and the level of competitive capabilities, a relationship we did in fact observe when we plotted these variables together. It was only the firms with an intermediate level of capabilities that revealed the high levels of participation in the regional institution.

While our research has deepened our understanding of the links between networks and capabilities and the role of regional institutions a number of further questions arise. One fruitful area of inquiry is to tease out the longitudinal dynamics of how firms' networks evolve and change in response to emerging competitive challenges and opportunities. As a firm's competitive environment changes, to what extent does inertia constrain a firm's ability to reconfigure its pattern of interfirm linkages? Further, our understanding of the processes of cooperative competition in geographical clusters could greatly benefit from a more fine-grained analysis of the mix of cooperation and competition in networks. While a popular idea (Hamel, 1991), the balance between cooperation and competition warrants greater research attention, particularly in the network context.

Future research could also consider additional network sources of firm heterogeneity, such as the interaction of multiplex relationships, with the aim of extending the literature on embeddedness in more tangible and empirically founded ways.

Concluding Remarks

Our research contributes to a better understanding of the role of firms' network of bridging ties in the development of competitive capabilities and participation in regional institutions. We take an initial step in the direction of examining economic action in geographical clusters by demonstrating that *firm differences* exist in terms of network structure and the degree of firms' participation in regional networks, and that these differences *do matter* for firms' acquisition of competitive capabilities.

Figure 1 - Illustration of Weak Ties and Structural Holes

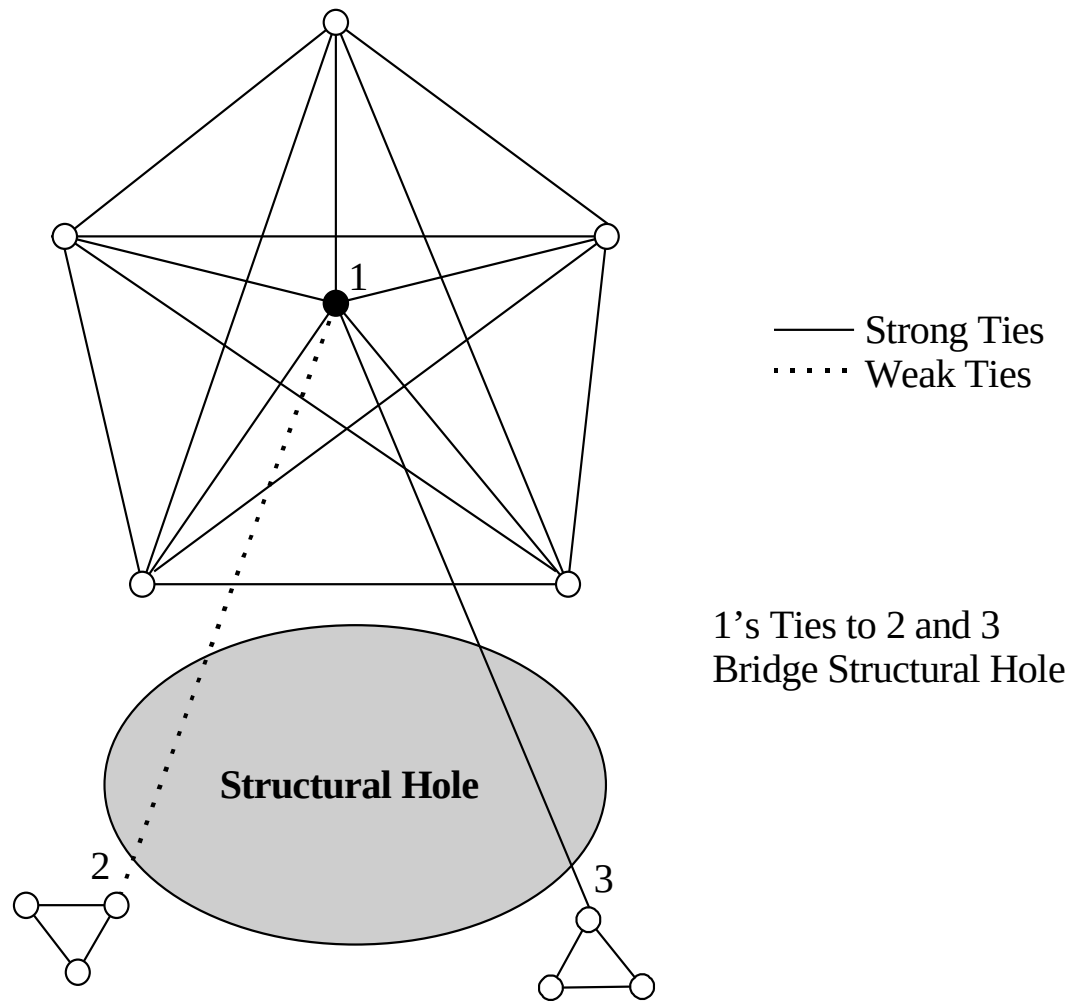


Table 1 - Measurement Instruments

Measurement Items	Scales	Internal Consistency Reliability (α)
Pollution Prevention Capabilities 1. Substitute less hazardous raw materials for more hazardous ones 2. Offer new products/services because of low waste disposal costs 3. Discontinue products/services high in environmental management costs	1 = We know little about this practice 2 = We know about this practice, but do <i>NOT</i> do it 3 = We have considered doing this 4 = We decided <i>NOT</i> to do this after considering it 5 = We do this from time to time 6 = We do this most of the time 7 = We do this all of the time	.72
Competitive Scanning Capabilities 1. Monitor your competitors' strategies and tactics 2. Search for information about which customers your competitors supply 3. Collect information about your competitors' market share	(same as above)	.81
Quality Management Capabilities 1. Collect data on your company's production process variations 2. Provide charts and graphs to production employees reporting defect rates 3. Conduct experiments to isolate causes of defects	(same as above)	.61

Table 1 (continued) - Measurement Instruments

Measurement Items	Scales	Internal Consistency Reliability (α)
Participation 1. Obtain on-site assistance at your company from [name of center] 2. Select/install new equipment or computer systems with [name of center] 3. Participate in user groups or networks organized by [name of center]	1 = We know little about this service 2 = We know about this service, but do <i>NOT</i> do it 3 = We considered doing this 4 = We decided <i>NOT</i> to do this after considering it 5 = We did this once 6 = We did this a couple of times 7 = We did this several times	.75
Non-Redundancy 1a. Please write the initials of the five most important people <i>not employed by your company</i> that you rely on for advice about managing your business. 1b. Now, using the table provided indicate if these people know each other. If so, circle 'Y' for yes.	(not applicable)	(not applicable)
Infrequency of Interaction 1. Please tell us approximately how many conversations per month (on average) you have with each of your advisors about your business.	(not applicable)	(not applicable)
Geographic Dispersion 1. What is the travel time by car to each advisor's office?	(not applicable)	(not applicable)

Table 2 - Descriptive statistics and zero-order correlations among constructs

	Mean	S.D.	1	2	3	4	5	6	7	8
1 Pollution Prevention	3.831	1.778	1.000							
2 Competitive Scanning	3.363	1.551	0.260**	1.000						
3 Quality Management	4.510	1.501	0.295**	0.323**	1.000					
4 Participation	2.131	1.405	0.132*	0.098	0.215**	1.000				
5 Non-Redundancy	0.990	0.629	0.130*	0.135*	0.032	-0.121*	1.000			
6 Infrequency of Interaction	0.626	0.335	-0.014	-0.115	0.016	-0.0583	-0.071	1.000		
7 Geographic Dispersion	1.095	0.886	-0.021	0.126*	0.083	0.007	0.091	0.031	1.000	
8 Firm Size	0.000	1.000	0.020	0.176**	0.264**	0.032	-0.038	-0.064	0.026	1.000

Table 3 - Discriminant Analysis

Model	d.f.	χ^2	$\Delta\chi^2$	p-value of $\Delta\chi^2$
Independent (exogenous) Latent Variables				
All ϕ s left free	3	1.98	--	--
ϕ_{12} fixed to 1.0 (Non-Redundancy & Infrequency of Interaction)	4	70.48	68.50	.000
ϕ_{13} fixed to 1.0 (Non-Redundancy & Geographic Dispersion)	4	43.72	41.74	.000
ϕ_{23} fixed to 1.0 (Infrequency of Interaction & Geographic Dispersion)	4	52.77	50.79	.000
Dependent (endogenous) Latent Variables				
All ψ s left free	48	64.80	--	--
ψ_{23} fixed to 1.0 (Pollution Prevention & Quality Management)	49	134.49	69.69	.000
ψ_{24} fixed to 1.0 (Pollution Prevention & Strategy Formulation)	49	136.70	71.90	.000
ψ_{34} fixed to 1.0 (Quality Management & Strategy Formulation)	49	130.40	65.60	.000

Figure 2 - Bridging Ties Structural Equation Model

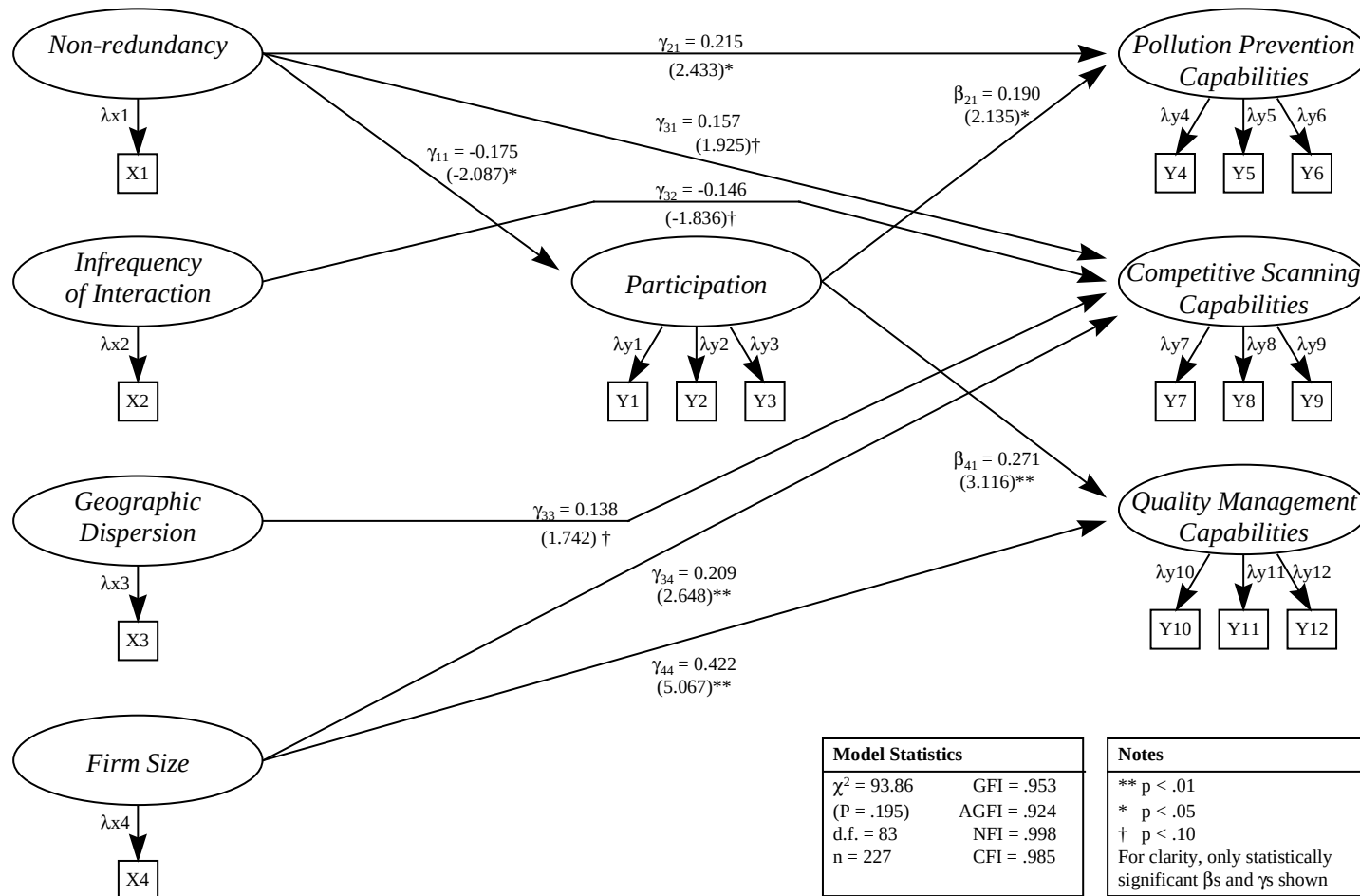


Table 4 - Parameter Estimates for Structural Model (Figure 2)

Parameter	Standardized Solution	t-value
β_{21}	0.190	2.135*
β_{31}	0.126	1.546
β_{41}	0.271	3.116**
γ_{11}	-0.175	-2.087*
γ_{21}	0.215	2.433*
γ_{31}	0.157	1.925†
γ_{41}	0.066	0.785
γ_{12}	-0.083	-1.000
γ_{22}	-0.009	-0.106
γ_{32}	-0.146	-1.836†
γ_{42}	0.019	0.228
γ_{13}	0.031	0.376
γ_{23}	-0.061	-0.714
γ_{33}	0.138	1.742†
γ_{43}	0.060	0.733
γ_{14}	0.040	0.485
γ_{24}	-0.026	-0.310
γ_{34}	0.209	2.648**
γ_{44}	0.422	5.067**
λx_1	0.900	(Fixed Parameter)
λx_2	0.900	(Fixed Parameter)
λx_3	0.900	(Fixed Parameter)
λx_4	0.900	(Fixed Parameter)
λy_1	0.817	(Fixed Parameter)
λy_2	0.763	8.536**
λy_3	0.593	7.665**
λy_4	0.776	(Fixed Parameter)
λy_5	0.662	7.355**
λy_6	0.598	7.000**
λy_7	0.828	(Fixed Parameter)
λy_8	0.716	9.961**
λy_9	0.755	10.320**
λy_{10}	0.801	(Fixed Parameter)
λy_{11}	0.476	5.490**
λy_{12}	0.485	5.565**

**P < .01, * P < .05, † P < .10

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